Shanmukh Upadhyayula

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Education

University of Illinois Urbana-Champaign GPA: 3.95/4.0

Master of Computer Science (MCS), Planned *August 2027 – May 2028*

Bachelor of Science in Computer Science & Geographic Information Science *August 2024 – May 2027*

- Involvement: ACM SIG Project Lead for HCI and AI, Virtual Reality Physics Simulation Developer
- Relevant Coursework: Data Structures, Algorithms & Models of Computation, Computer Systems, Database Management in the Cloud, Applied Linear Algebra, Spatial Analysis

Experience

UIUC CyberGIS Lab Urbana, IL

Machine Learning Engineering Intern *September 2026 – Present*

- Benchmarking the lab's geospatial AI agent on all 50 GeoAnalystBench spatial analysis tasks, showing it matches published LLMs on code but over-plans on 43 of 48 tasks
- Uncovering 4 feature-breaking bugs by user-testing the agent on search, projections, rasters, and adversarial prompts
- Reworking the agent's tool selection, search indexing, and code execution to target over-planning and bugs from testing

Semantic Segmentation Failure Analysis Urbana, IL

Undergraduate Researcher | shanmukhupad.github.io/portfolio/seg.pdf *May 2026 – August 2026*

- Developed a label-free way to map where a DeepLabV3-ResNet50 land-cover model is likely wrong across full-size aerial rasters, by scanning them tile by tile with the model's own confidence
- Discovered the model is about half as accurate along object edges by validating the error maps on 4 benchmark datasets, and presented the work in an IEEE-format paper and a CyberGIS talk

Chicago Food Inspection Model Bias Urbana, IL

Undergraduate Researcher | shanmukhupad.github.io/portfolio/bias_reduction.html *May 2026 – August 2026*

- Cut the largest recall gap between Chicago neighborhoods by 99% by writing a custom Group-DRO XGBoost objective with a hand-derived gradient and Hessian
- Predicted which bias fix would work by tracing the gap to a ranking deficit rather than a threshold artifact, using per-group ROC AUC across 8,312 inspections

ACM SIG-AIDA Urbana, IL

Project Lead | github.com/ShanmukhUpad/Pathway-To-Improved-Cities *February 2026 – May 2026*

- Led an 8-person team building a Streamlit and Mapbox GL dashboard that turns any uploaded urban dataset into on-demand maps and Random Forest and Gradient Boosting predictions
- Made model errors and spatial patterns easy to spot across 6 urban domains by engineering automated data validation, Moran's I and LISA hot-spot analysis, and actual vs. predicted map views

Projects

Spatial Autocorrelation of AI Data Centers | *Python, PySAL* March 2026 – May 2026

- Revealed that 5 significant data center impact effects vanish under a spatial-lag model compared with OLS
- Detected a hidden Northern Virginia electricity-price hot spot by mapping local clusters with LISA

Chicago Walkability Analysis | *R* | shanmukhupad.github.io/chicago-walkability October 2025 – April 2026

- Integrated a walkability measure into Chicago's ChiVes platform with an R regression of urban form ($R^2 = 0.85$)
- Demonstrated that mixed land use and street connectivity, not population density, drive walkability by mapping spatial clusters with Moran's I, and presented the findings at the UIUC Undergraduate Research Symposium

Urban AR GIS: 3D City Viewer | *Python, Shapely, pyproj* | github.com/ShanmukhUpad/urban-ar-gis March 2026

- Turned OpenStreetMap data for any latitude and longitude into a navigable 3D city by projecting building footprints to local meters with pyproj and triangulating them into 3D models
- Rendered the city over high-resolution satellite tiles at 60 FPS with ModernGL, letting users inspect building metadata and move buildings with hand gestures tracked at about 30 FPS

Technical Skills

Languages: Python, C++, C, Java, TypeScript, JavaScript, SQL, R
Geospatial: GeoPandas, Shapely, pyproj, PySAL, sf, ArcGIS Pro, ArcPy, LiDAR, DEMs, georeferencing
Tools: PyTorch, NumPy, Pandas, OpenCV, Git, Docker, Google Cloud, Firestore, MongoDB